



Introduction

A **differential equation** is an equation that contains one or more terms involving derivatives of one variable (the dependent variable, y) with respect to another variable (the independent variable, x). In other words, the variable is called dependent if a derivative of that variable occurs. Usually, differential equations are used as models to approximate real-world problems.

Examples of Differential Equations:

- a. $\frac{dy}{dx} = \sin x$ *Excel Review Center*
 b. $(2xy + x^2) dx + 2y dy = 0$
 c. $\left(\frac{d^3y}{dx^3}\right)^2 - x\left(\frac{d^2y}{dx^2}\right)^3 + x^3 = 0$
 d. $(x - 1)y'' + xy' + y = 0$
 e. $\frac{\partial x}{\partial y} + 2y \frac{\partial x}{\partial z} = 3$
 f. $\frac{d^2y}{dt^2} + \frac{d^2x}{dt^2} = x$

The solutions of differential equations are functions and not just numbers like the algebraic equations.

Example:

The differential equation $\frac{dy}{dx} = 2x$ has for its solution $y = \int 2x dx = x^2 + C$, where C is an arbitrary constant.

Types of Differential Equations

1. **Ordinary Differential Equation** – an equation containing only one independent variable, thus having only **ordinary derivatives** in the equation. *Excel Review Center*

Examples a, b, c, d and f above are examples of ordinary differential equations

2. **Partial Differential Equation** – an equation containing two or more independent variables, thus having **partial derivatives** in the equation.

Example e above is an example of partial differential equation.

Order of Differential Equations

The order of a DE is the **order of the highest ordered derivative** which appears in the equation.

Examples: *Excel Review Center*

First Order: $\frac{dy}{dx} = 2x^2 + 5x + 3$

Second Order: $\frac{d^2y}{dx^2} = 5x^2 + 10x + 3$

Third Order: $\frac{d^3y}{dx^3} = 3x^2 + 6x + 2$

Degree of Differential Equations

The degree of a differential equation whose terms are polynomials in the derivatives is defined as the **highest power of the highest order derivative**.

Examples: *Excel Review Center*

First Degree: $x \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 = 15$

Second Degree: $x \left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^3 = 15$

Types of Solutions of Differential Equations

1. **General solution** – the solution has at least one arbitrary constant. *Excel Review Center*

Example: $x^3 - 3y = C$

where: C = arbitrary constant

2. **Particular solution** – the solution that has **no** arbitrary constant.

Example: $2xy + y^2 - 2 = 0$

Solutions to First Order DE

1. **Variable separable type**

If a first order differential equation is of such a form that all the occurrences of one variable may be moved to one side of the equality and all occurrences of the other variable to the other side, the equation is called separable. Such equation may be solved by integrating separate expressions.

Standard form: *Excel Review Center*

$$P(x)dx + Q(y)dy = 0$$

General solution:

$$\int P(x)dx + \int Q(y)dy = C$$

2. **Homogeneous type**

Standard form:

$$M(x,y)dx + N(x,y)dy = 0$$

General solution:

Substitute $y = vx$ or $x = vy$ and the resulting DE becomes a variable separable type

Note:

First-order differential equations are not always separable. Homogeneous differential equations, however, may be transformed into separable equations by the substitution of a variable.

An expression is said to be homogeneous if all terms have the same degree. The term “homogeneous” is also used to indicate that the right-hand member of a linear differential equation is 0

Test for homogeneity of degree n :

Let: $f(x,y) = M(x,y) dx + N(x,y) dy$
 $f(rx, ry) = r^n f(x,y)$

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